DONOR BEHAVIOR ANALYSIS & FORECASTING

DRIVING IMPACT WITH DATA: INSIGHTS FOR NOT-FOR-PROFITS

BY KAGNA EM

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PROJECT OVERVIEW

This Data Insights Project, completed in mid 2025, focused on analyzing supporter engagement trends to generate actionable reports. The insights enabled the team to target the right supporters at the right time, significantly enhancing outreach effectiveness.

Data Source: For confidentiality, Mockup data was created based on actual datasets live-streamed from Microsoft Dynamics 365 CRM, supplemented with external sources such as Stats NZ and OneRoof, covering five fiscal years.

Due to the volume and complexity of data, Microsoft Excel was insufficient for effective reporting and visualization.

Power BI is utilized to manage and visualize large-scale data in real time.

Tools & Technologies: MS CRM Dynamic 365, **Power BI (With Fabric)**, **DAX**, **Power Query**, **API integration**, and **Python** for data modeling, transformation, and automation.

METHODOLOGY

MS DYNAMIC 365 CRM DATA MANAGEMENT

DYNAMIC CONNECTION & DATA PIPE LINE

DEVELOPING DATA MODEL IN POWER BI

REPORT DASHBOARD

DATA FLOW/PIPELINE (ELT)

This project focused on donor information spanning the past five fiscal years.With approximately one million donation records, extracting, loading, and transforming data from the CRM to Power BI posed significant challenges if not handled efficiently.

To address these challenges, a data pipeline and dataflow were implemented to reduce loading times and minimize errors.

Power BI Online, with Microsoft Fabric, was used as a centralized data preparation layer through Dataflows.

Power BI Desktop was used for data modeling and report creation, consuming the cleaned and transformed data.

A well-defined workflow was established to ensure seamless integration between the CRM, Power BI Online, and Power BI Desktop.

DATA FLOW/PIPELINE (ELT)

- **Below is the process used to create the Data Pipeline with Microsoft Fabric (ELT approach) in the Microsoft 365 environment.**
- **The process began in Power BI Service (Online) within an environment where Microsoft Fabric was enabled**

1. Create a Dataflow Gen2
in a Fabric-enabled workspace
→ **Extract & Load:** Connect to CRM tables via wizard or script (here I use script to avoid loading error due complex data structure, load raw data into **Lakehouse**
→ **Transform:** Clean and shape data using Power Query before or after loading

2. Schedule Dataflow Refresh
→ Configure auto-refresh (weekly)
→ Ensure credentials are correctly set

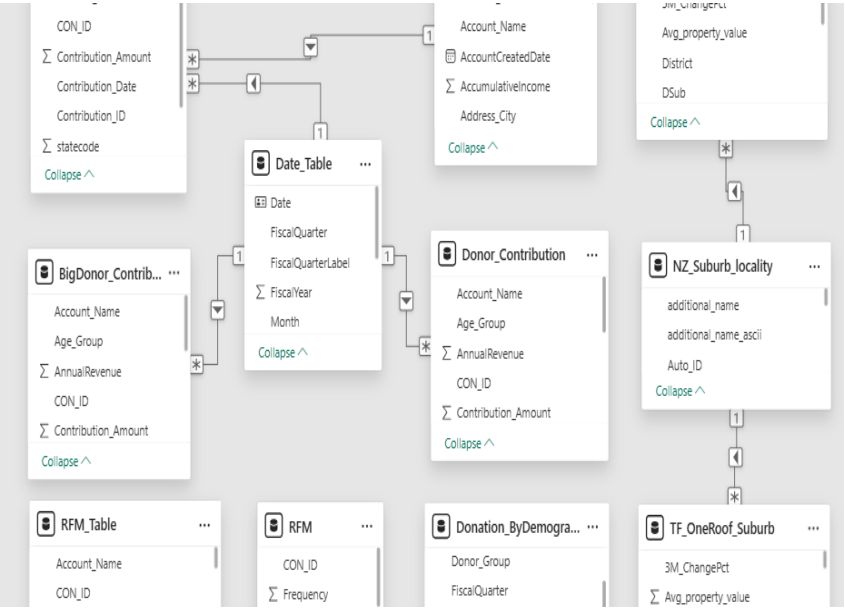
3. Lakehouse stores structured data
→ Tables accessible via OneLake SQL endpoint or Lakehouse connector

```
),

// Select only required columns
SelectedColumns = Table.SelectColumns(FilteredRows, {
    "tf_conid", "name", "totalamount", "statuscode", "statec
}),

// Reorder columns
#"Reordered Columns" = Table.ReorderColumns(SelectedColumns,
    "datedelivered", "tf_conid", "name", "totalamount", "sta
}),

// Sort by delivery date
```

Loading data to Power BI for Data Modeling and Visualization

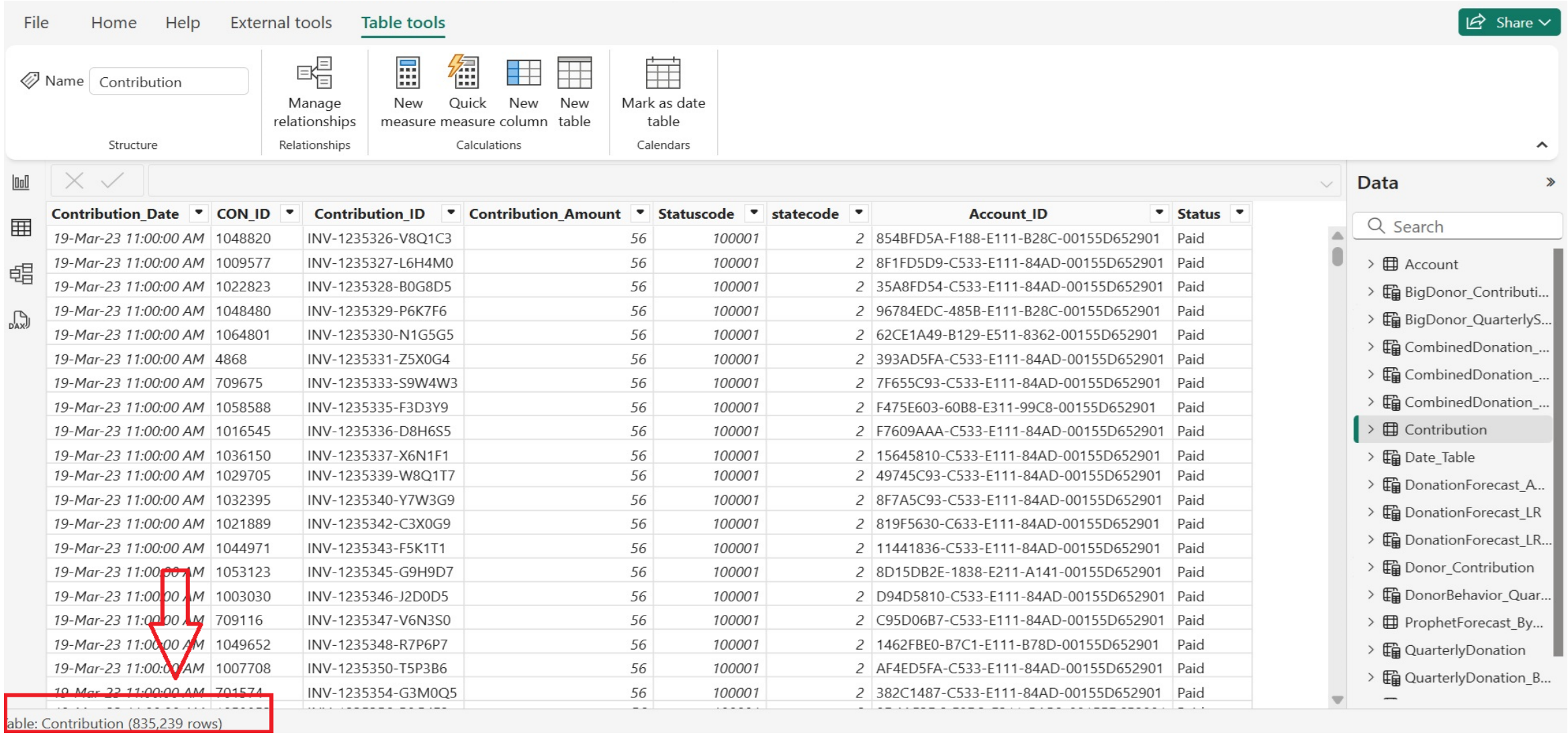
Power BI Desktop was used to load data from the created pipeline for data modeling, developing advanced DAX measures and table calculations, and building interactive visuals and dashboards.

After getting all relevant data into Power BI, Data modeling is another critical step. Modeling data in Power BI involves structuring relationships between tables, creating DAX measures, and optimizing data for analysis. A well-designed model improves performance, ensures consistency, and enables interactive, insightful reporting.



LARGE DATASET TRANSFORMATION

This is how the cleaned and structured data tables appear. After filtering, over 80,000 records were successfully transformed in minimal time and are ready for use.



Contribution_Date	CON_ID	Contribution_ID	Contribution_Amount	Statuscode	statecode	Account_ID	Status
19-Mar-23 11:00:00 AM	1048820	INV-1235326-V8Q1C3	56	100001	2	854BFD5A-F188-E111-B28C-00155D652901	Paid
19-Mar-23 11:00:00 AM	1009577	INV-1235327-L6H4M0	56	100001	2	8F1FD5D9-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1022823	INV-1235328-B0G8D5	56	100001	2	35A8FD54-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1048480	INV-1235329-P6K7F6	56	100001	2	96784EDC-485B-E111-B28C-00155D652901	Paid
19-Mar-23 11:00:00 AM	1064801	INV-1235330-N1G5G5	56	100001	2	62CE1A49-B129-E511-8362-00155D652901	Paid
19-Mar-23 11:00:00 AM	4868	INV-1235331-Z5X0G4	56	100001	2	393AD5FA-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	709675	INV-1235333-S9W4W3	56	100001	2	7F655C93-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1058588	INV-1235335-F3D3Y9	56	100001	2	F475E603-60B8-E311-99C8-00155D652901	Paid
19-Mar-23 11:00:00 AM	1016545	INV-1235336-D8H6S5	56	100001	2	F7609AAA-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1036150	INV-1235337-X6N1F1	56	100001	2	15645810-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1029705	INV-1235339-W8Q1T7	56	100001	2	49745C93-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1032395	INV-1235340-Y7W3G9	56	100001	2	8F7A5C93-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1021889	INV-1235342-C3X0G9	56	100001	2	819F5630-C633-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1044971	INV-1235343-F5K1T1	56	100001	2	11441836-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	1053123	INV-1235345-G9H9D7	56	100001	2	8D15DB2E-1838-E211-A141-00155D652901	Paid
19-Mar-23 11:00:00 AM	1003030	INV-1235346-J2D0D5	56	100001	2	D94D5810-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	709116	INV-1235347-V6N3S0	56	100001	2	C95D06B7-C533-E111-84AD-00155D652901	Paid
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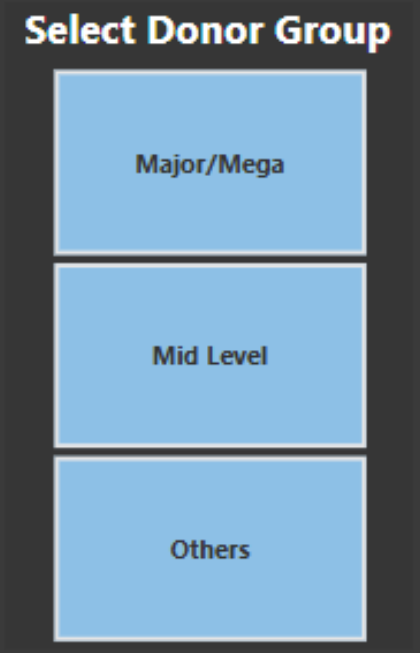
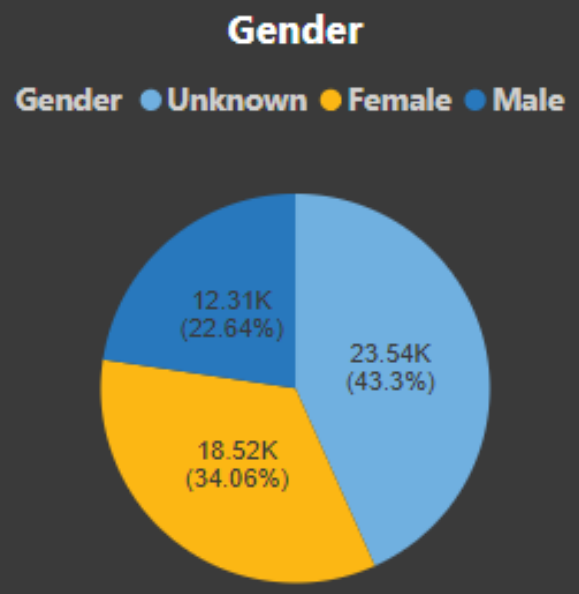
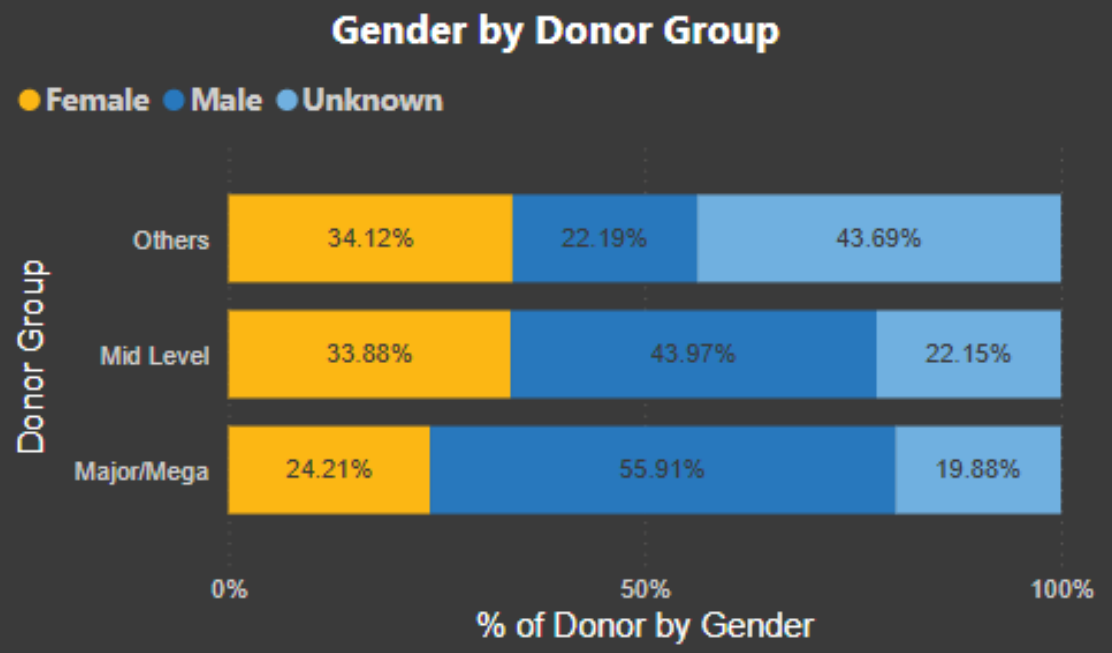
19-Mar-23 11:00:00 AM	1007708	INV-1235350-T5P3B6	56	100001	2	AF4ED5FA-C533-E111-84AD-00155D652901	Paid
19-Mar-23 11:00:00 AM	701574	INV-1235354-G3M0Q5	56	100001	2	382C1487-C533-E111-84AD-00155D652901	Paid

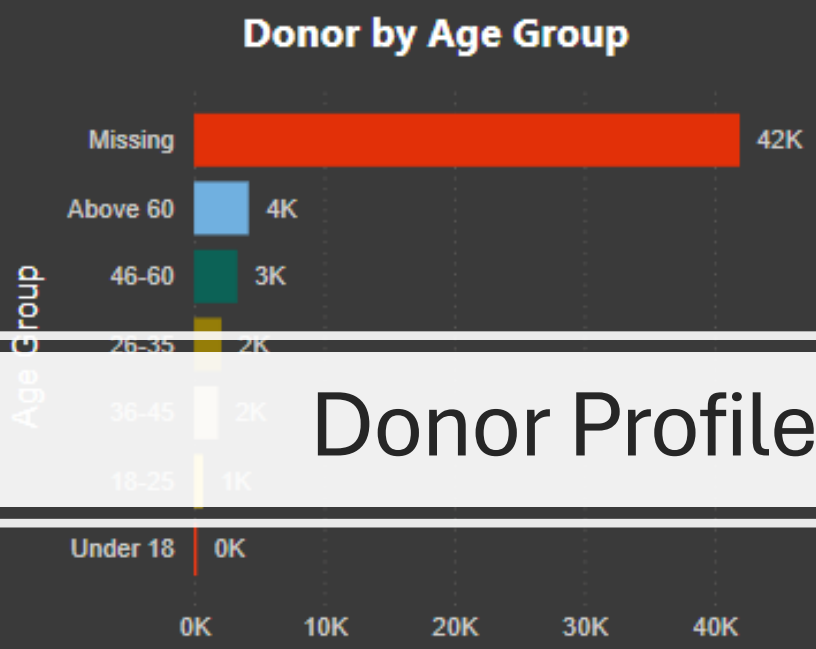
REPORT DASHBOARD

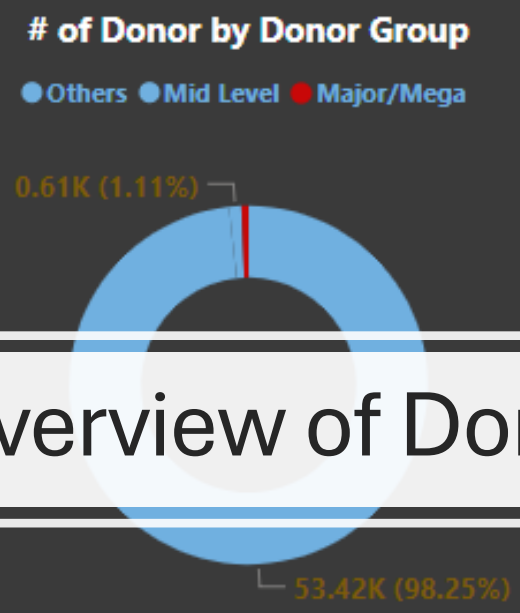
Donor Profile – Overview of Donor Characteristics

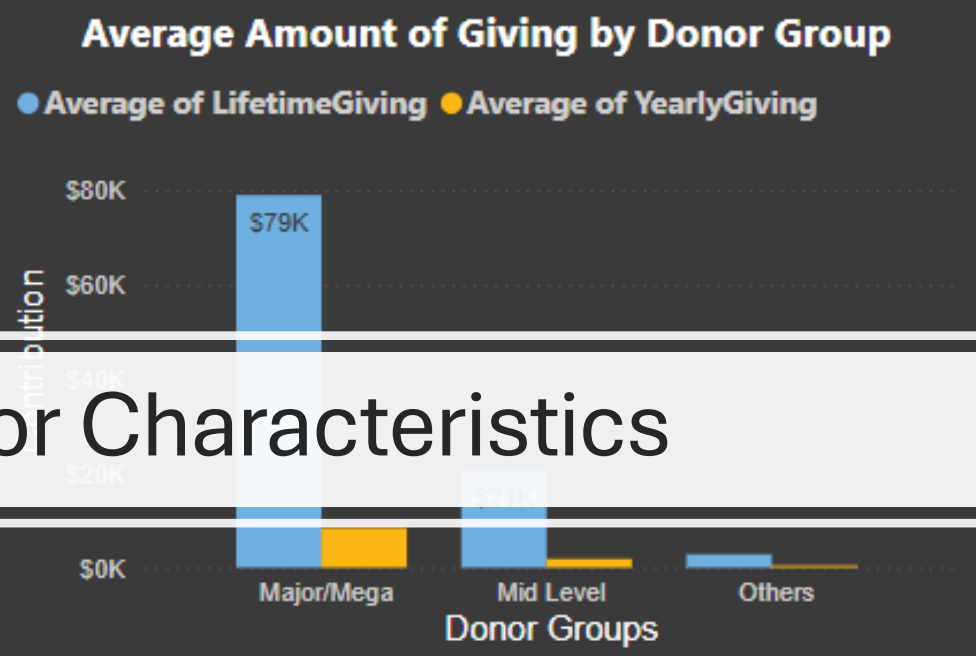




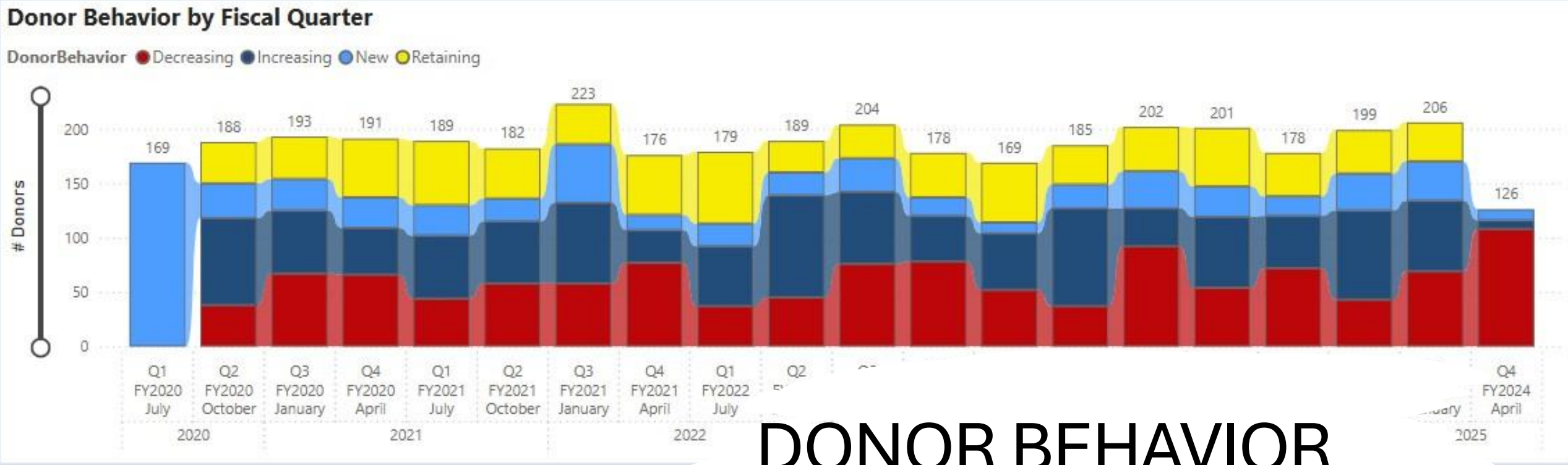




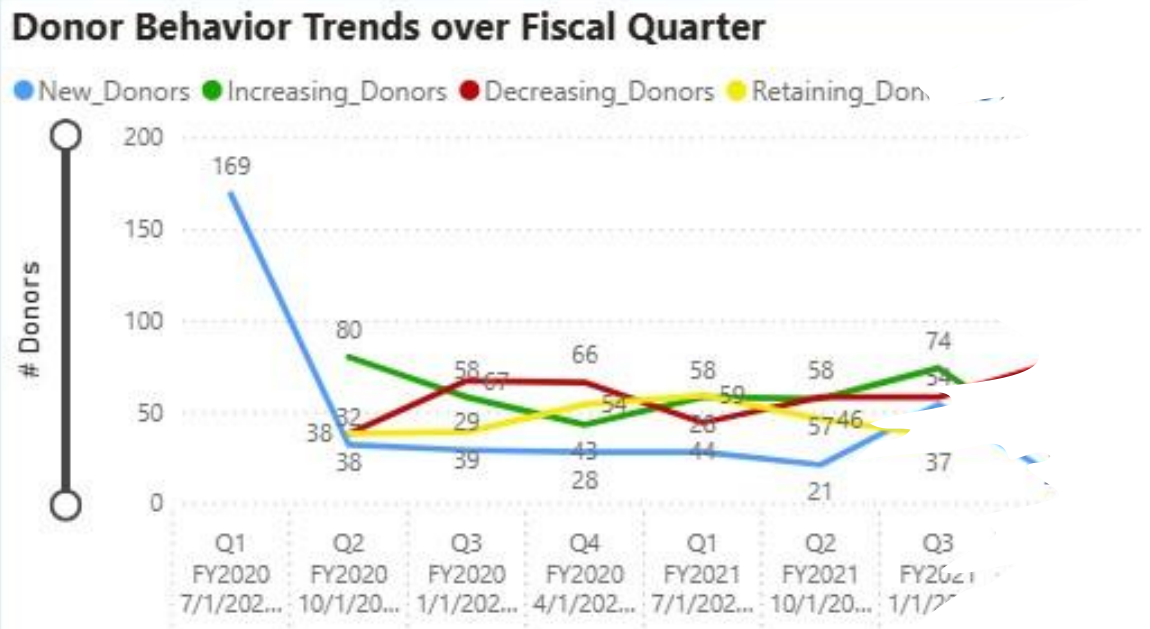




Donation Trend – Overview of Donor Contributions



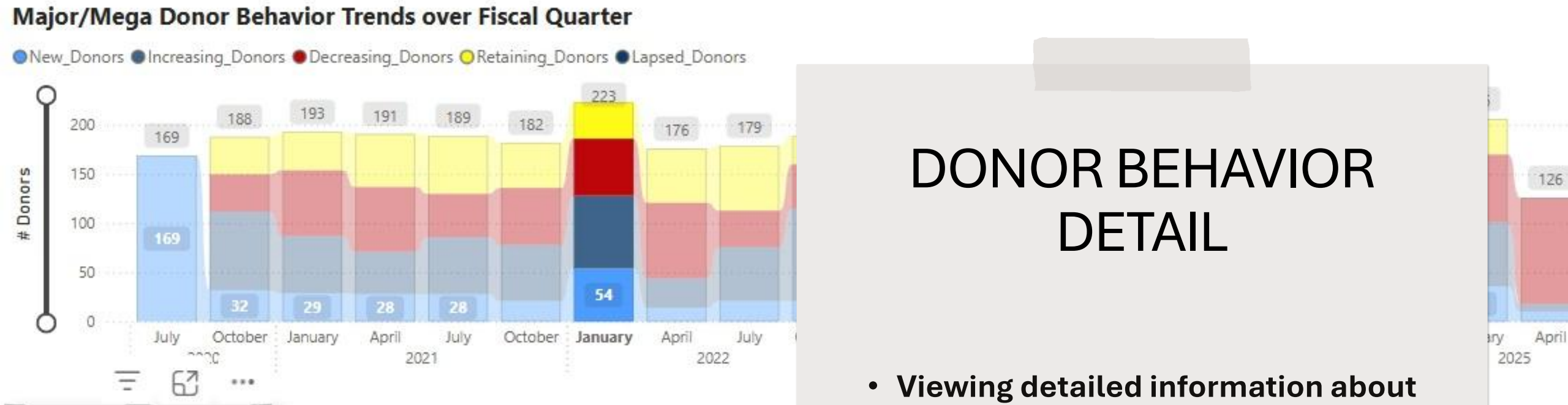
DONOR BEHAVIOR ANALYSIS



- **Spanning five fiscal years, the bar chart and line chart illustrate the quarterly total number of donors categorized by their donation behaviors: new, increasing, decreasing, retained, or lapsed.**

DONOR BEHAVIOR DETAIL

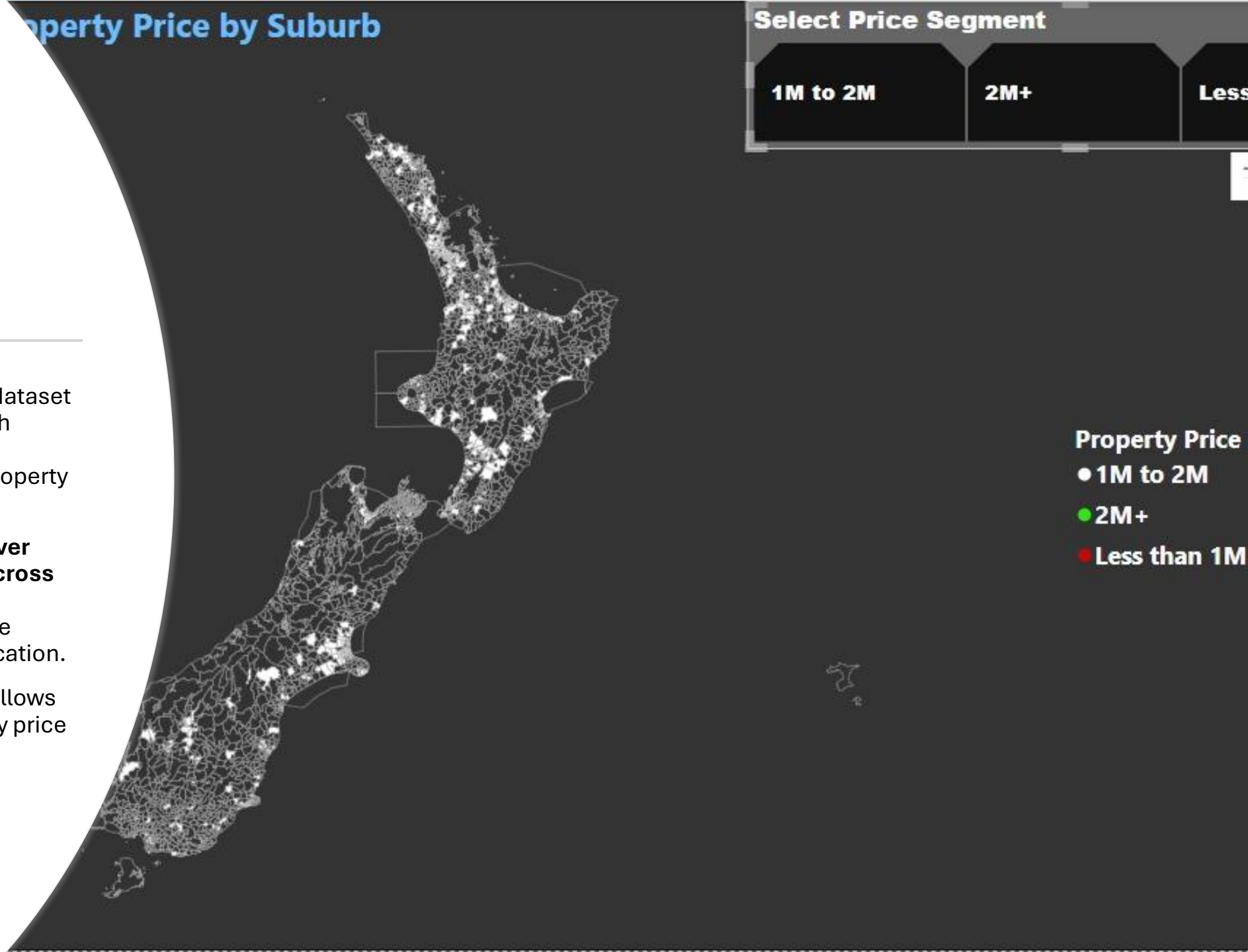
- **Viewing detailed information about donors by behavior within each fiscal quarter adds further insight. Clicking on a bar in the chart reveals donor details below, while the slicer in the bottom left corner allows filtering by specific donation behavior.**

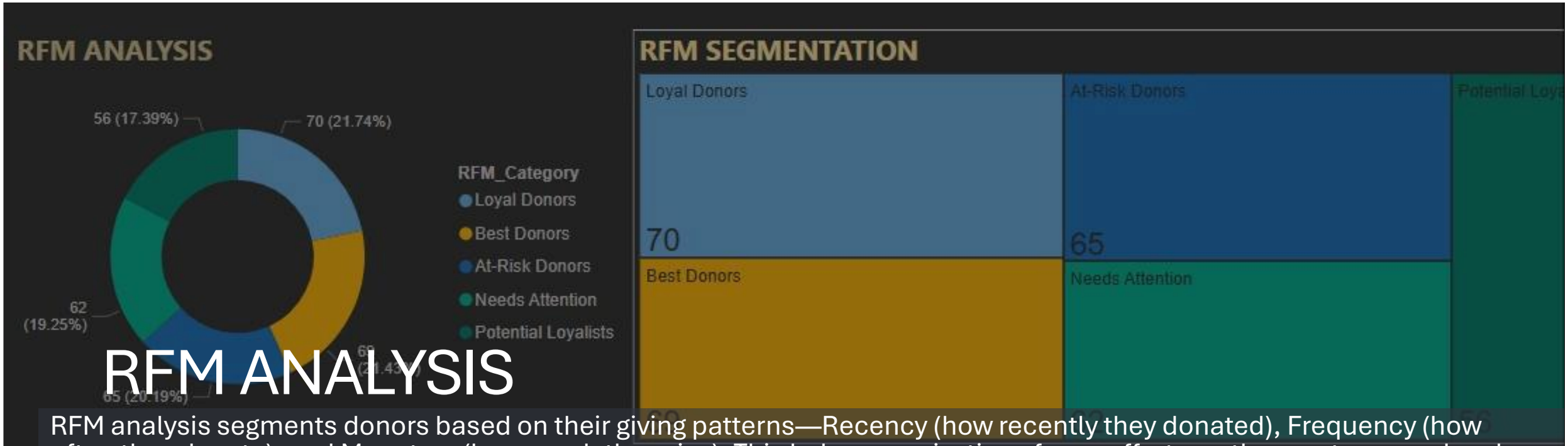


ID	Account_Name			
AA001	Mr & Mrs AAA			
BB002	BBB Foundation			
CC003	CCC Limited			
		Q3 FY2021	Decreasing	
		Q3 FY2021	Decreasing	$1,444
		Q3 FY2021	Decreasing	$1,400
Total				$67,895.48

DONOR MAPPING

- The final product is a combined dataset integrating donor information with external data sources, including boundary maps from LINZ and property data from OneRoof.
- **This interactive map displays over 1000 suburb boundary layers across New Zealand**, visualizing donor information alongside the average property price in each donor's location.
- A **slicer in the top-right corner** allows users to filter the map by property price segments.





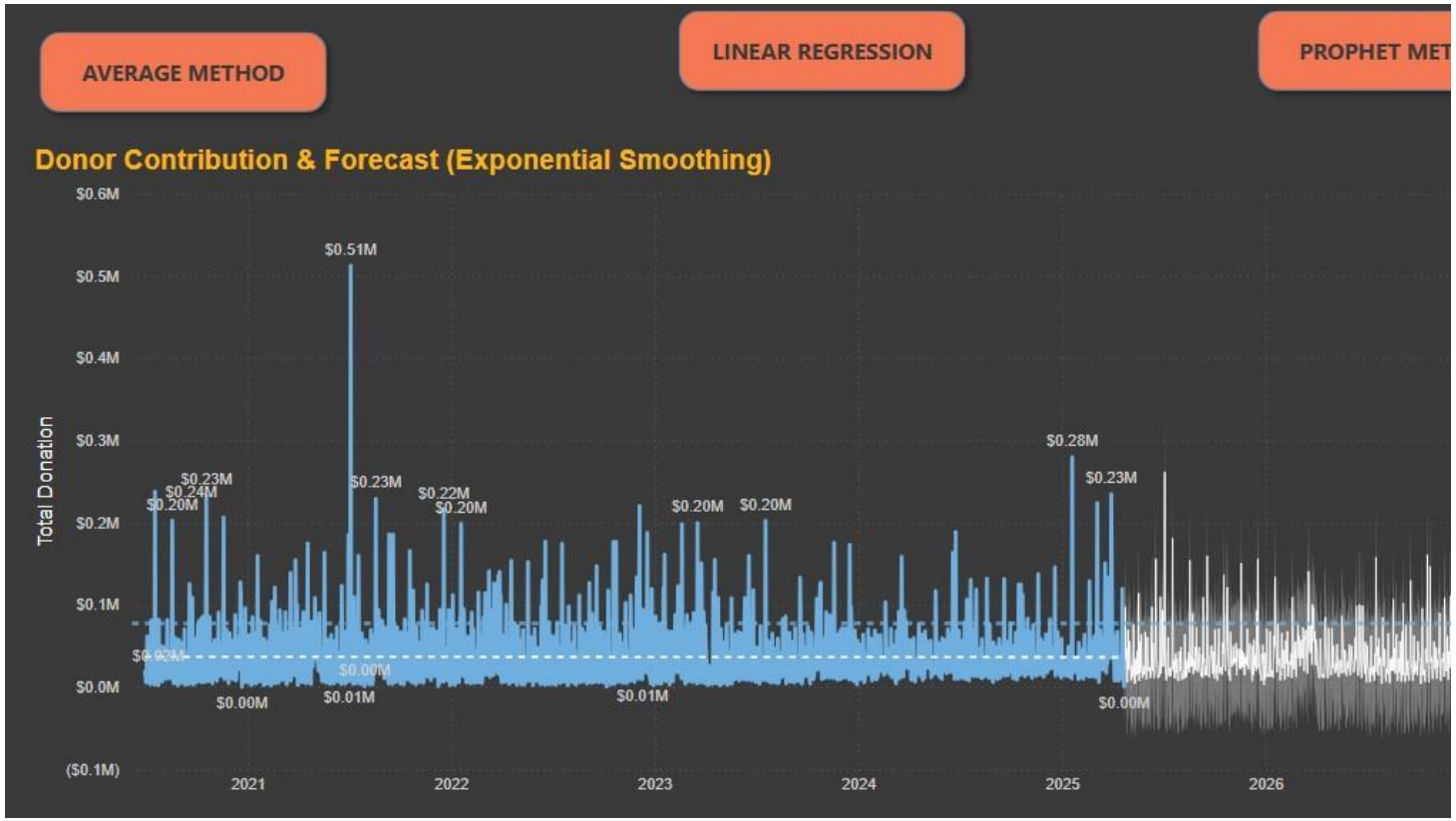
RFM ANALYSIS

RFM analysis segments donors based on their giving patterns—Recency (how recently they donated), Frequency (how often they donate), and Monetary (how much they give). This helps organizations focus efforts on the most engaged and valuable supporters, making it a powerful tool for optimizing outreach and increasing donor loyalty

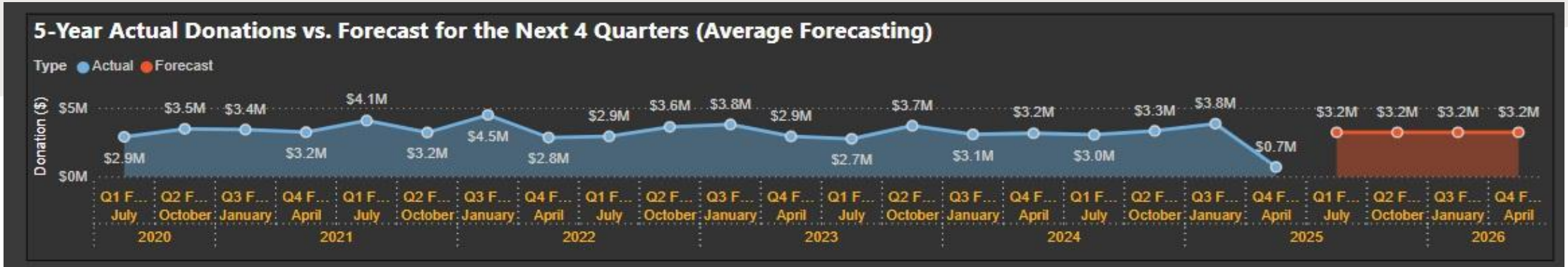
CON_ID	Account_Name	Donor_Group	Recency	Frequency	Monetary	RFM_Score	Recency_Score	Frequency_Score	Monetary_Score	RFM_Score	RFM_Ca
[redacted]	[redacted] Foundation	Major/Mega	164	1	$5,000	211	2	1	1	211	Needs
[redacted]	[redacted]	Major/Mega	47	4	$10,000	422	4	2	2	422	Loyal D
[redacted]	[redacted]	Major/Mega	67	4	$5,000	321	3	2	1	321	Potenti
[redacted]	[redacted]	Major/Mega	52	13	$11,090	432	4	3	2	432	Loyal D
[redacted]	[redacted] Trust	Major/Mega	223	5	$20,500	223	2	2	3	223	Needs
[redacted]	[redacted]	Major/Mega	55	6	$37,050	424	4	2	4	424	Loyal D
[redacted]	[redacted]	Major/Mega	203	3	$30,000	224	2	2	4	224	Needs

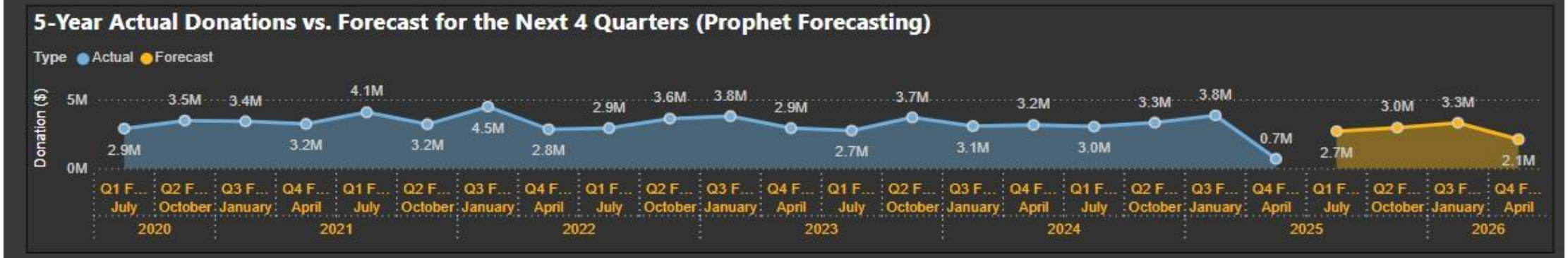
FORCASTING

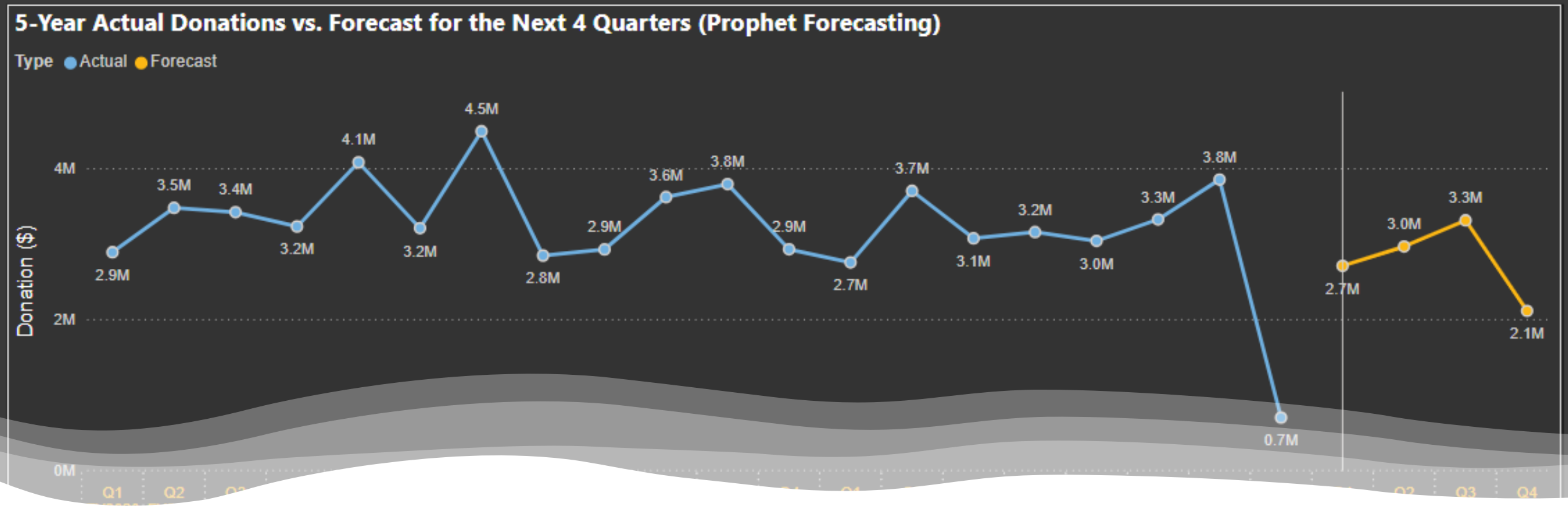
- This line chart is used for displaying actual donations over the past five fiscal years compared to forecasted donations for the next four quarters. See the next figure for a comparison among 3 different forecasting methods: Average, Linear Regression and Phophet



COMPARATIVE ANALYSIS OF THREE FORECAST MODELS







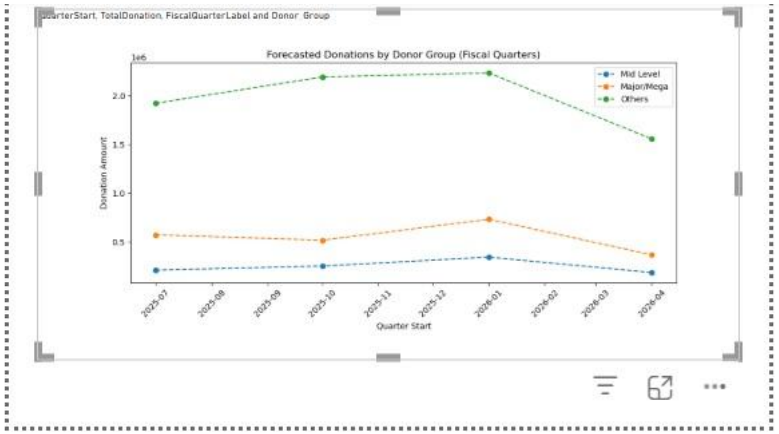
Prophet Model

Forecasting future donations using the Prophet model proved most effective given the data's complexity. This method is often considered effective—especially for business time series—because:

- Automatic Handling of Seasonality and Holidays – Captures recurring patterns and allows custom holiday inputs, making it ideal for business and donation trends.
- Robust and Flexible – Works well with missing data, outliers, and supports sudden trend changes using changepoints.
- Easy to Use and Interpretable – Requires minimal tuning, is open-source, and produces clear visual components for trend, seasonality, and forecasts.

Using Python Script

- **To use this model in Power BI, a Python script must be embedded. The script uses Prophet as the core forecasting algorithm and Matplotlib for data visualization. The process involves preparing the data in Power BI, passing it to Python for forecasting, and then returning the forecasted results back to Power BI for display in interactive visuals.**



Python script editor

⚠ Duplicate rows will be removed from the data.

```
warnings.filterwarnings("ignore", category=FutureWarning)

# Rename columns from Power BI input
df = dataset.rename(columns={
    dataset.columns[0]: 'ds',  # Date
    dataset.columns[1]: 'y',   # TotalDonation
    dataset.columns[2]: 'FiscalQuarterLabel',
    dataset.columns[3]: 'Donor_Group'
```

Using DAX

This analysis would not have been possible without the use of DAX for creating calculated tables and measures.

```
DonationForecast_LR_ByID =
GENERATE(
    SUMMARIZE(QuarterlyDonation_ByID, QuarterlyDonation_ByID[Donor_Group], QuarterlyDonation_ByID[CON_ID],
    QuarterlyDonation_ByID[Account_Name]),
    VAR ThisGroup = QuarterlyDonation_ByID[Donor_Group]
    VAR ThisDonor = QuarterlyDonation_ByID[CON_ID]
    VAR ThisName  = QuarterlyDonation_ByID[Account_Name]
    VAR GroupTable =
        FILTER(
            QuarterlyDonation_ByID,
            QuarterlyDonation_ByID[Donor_Group] = ThisGroup &&
            QuarterlyDonation_ByID[CON_ID] = ThisDonor &&
            QuarterlyDonation_ByID[Account_Name] = ThisName &&
            NOT ISBLANK(QuarterlyDonation_ByID[QuarterIndex]) &&
            NOT ISBLANK(QuarterlyDonation_ByID[TotalDonation])
        )
    VAR AvgX = AVERAGEX(GroupTable, QuarterlyDonation_ByID[QuarterIndex])
    VAR AvgY = AVERAGEX(GroupTable, QuarterlyDonation_ByID[TotalDonation])
    VAR Slope =
        DIVIDE(
            SUMX(
                GroupTable
```

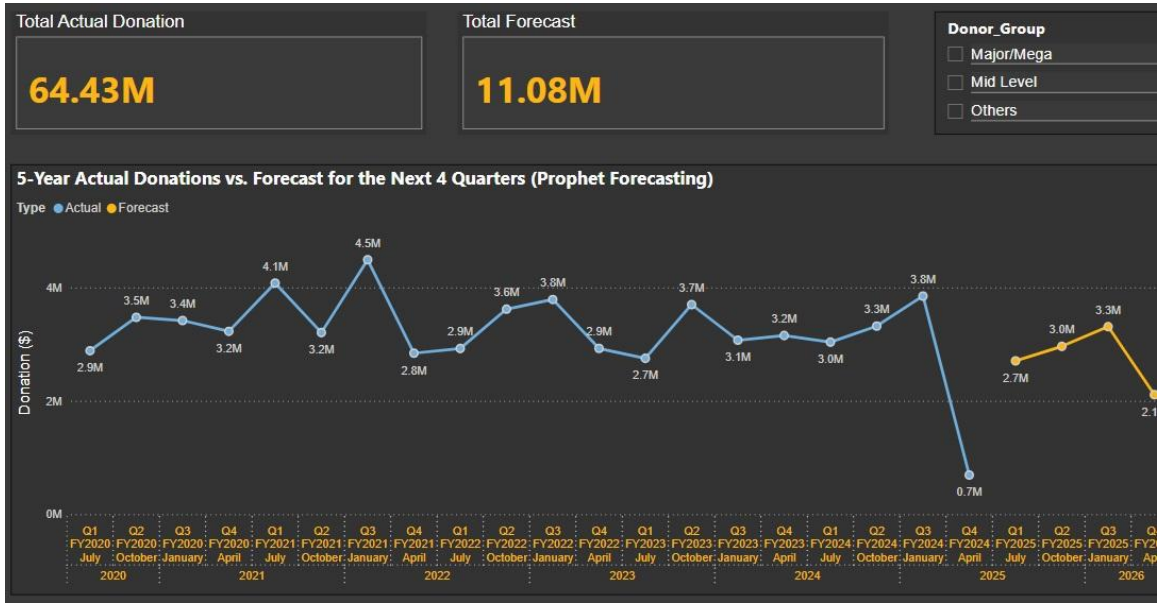
Donor_Group	CON_ID	Value	QuarterIndex	QuarterStart	ForecastDonation	FiscalQuarter	FiscalYear	FiscalQuarterLabel
Others	701430	1	209885	01-Jul-25 12:00:00 AM	$146.7886	Q1	2025	Q1 FY2025
Others	1005200	1	209885	01-Jul-25 12:00:00 AM	$146.7886	Q1	2025	Q1 FY2025

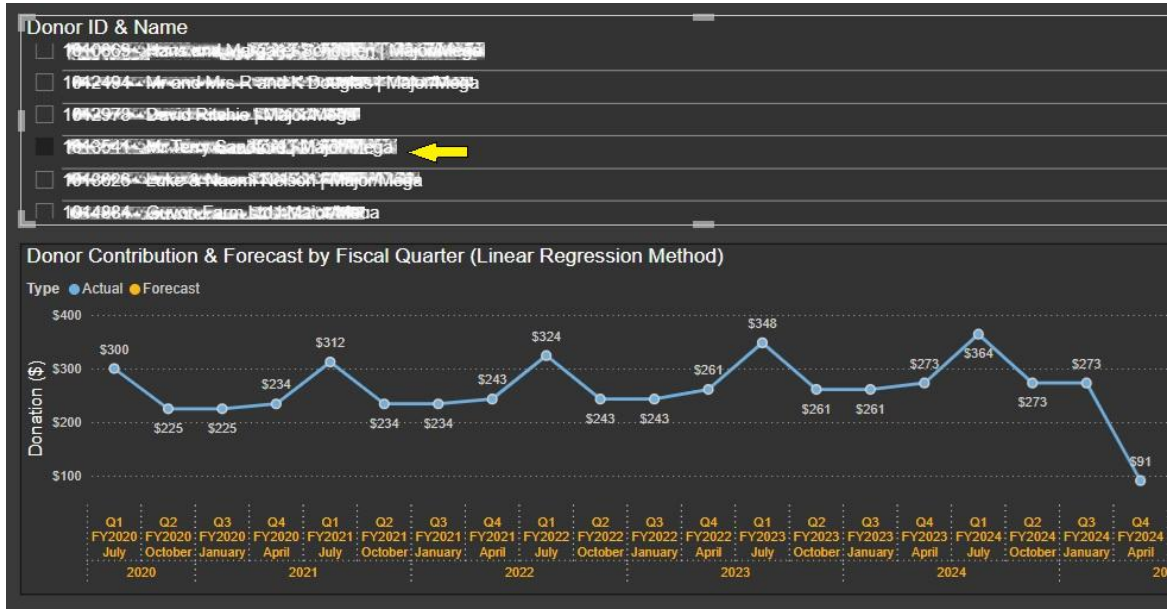
```
Date_Table =
VAR MinDate = MIN(Contribution[Contribution_Date])
VAR MaxDate = MAX(Contribution[Contribution_Date])
-- VAR MaxDateExtended = EDATE(MaxDate, 12) -- Optional: add 12 months for forecasting

RETURN
ADDCOLUMNS(
    CALENDAR(MinDate, MaxDate),
    "Year", YEAR([Date]),
    "Month", FORMAT([Date], "MMMM"),
    "MonthNum", MONTH([Date]),
    "MonthStart", DATE(YEAR([Date]), MONTH([Date]), 1),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "QuarterStart", DATE(YEAR([Date]), (QUARTER([Date]) - 1) * 3 + 1, 1),

    // Fiscal Year (starts in July)
    "FiscalYear",
        YEAR(EDATE([Date], -6)),  -- Shift 6 months back to align FY starting in July

    // Fiscal Quarter
    "FiscalQuarter",
        "Q" & SWITCH(
```

Date	Year	Month	MonthNum	MonthStart	Quarter	QuarterStart	FiscalYear	FiscalQuarter	FiscalQuarterLa
01-Jul-24 12:00:00 AM	2024	July	7	01-Jul-24 12:00:00 AM	Q3	01-Jul-24 12:00:00 AM	2024	Q1	Q1 FY2024
02-Jul-24 12:00:00 AM	2024	July	7	01-Jul-24 12:00:00 AM	Q3	01-Jul-24 12:00:00 AM	2024	Q1	Q1 FY2024

FORECAST AT THE DONOR/SEGMENT LEVEL

- The slicer is a valuable tool for enabling drill-down functionality—either by donor group (the left figure) or by individual donor account (the right figure).





SUMMARY

This Data Insights Project provides an overview of how large and complex CRM datasets can be effectively managed using modern business intelligence tools such as Power BI, API integration, DAX, and Python.

By enabling real-time interaction between CRM systems and decision-makers, the project supports data-driven strategies. Visualizations and techniques—including RFM analysis, donor profiling, contribution trends, donor behaviors, and donor mapping—deliver actionable insights that drive more targeted outreach and strategic decisions.

Most importantly, forecasting donation trends offers a high-level view to inform future planning and broader organizational goals

THANK YOU